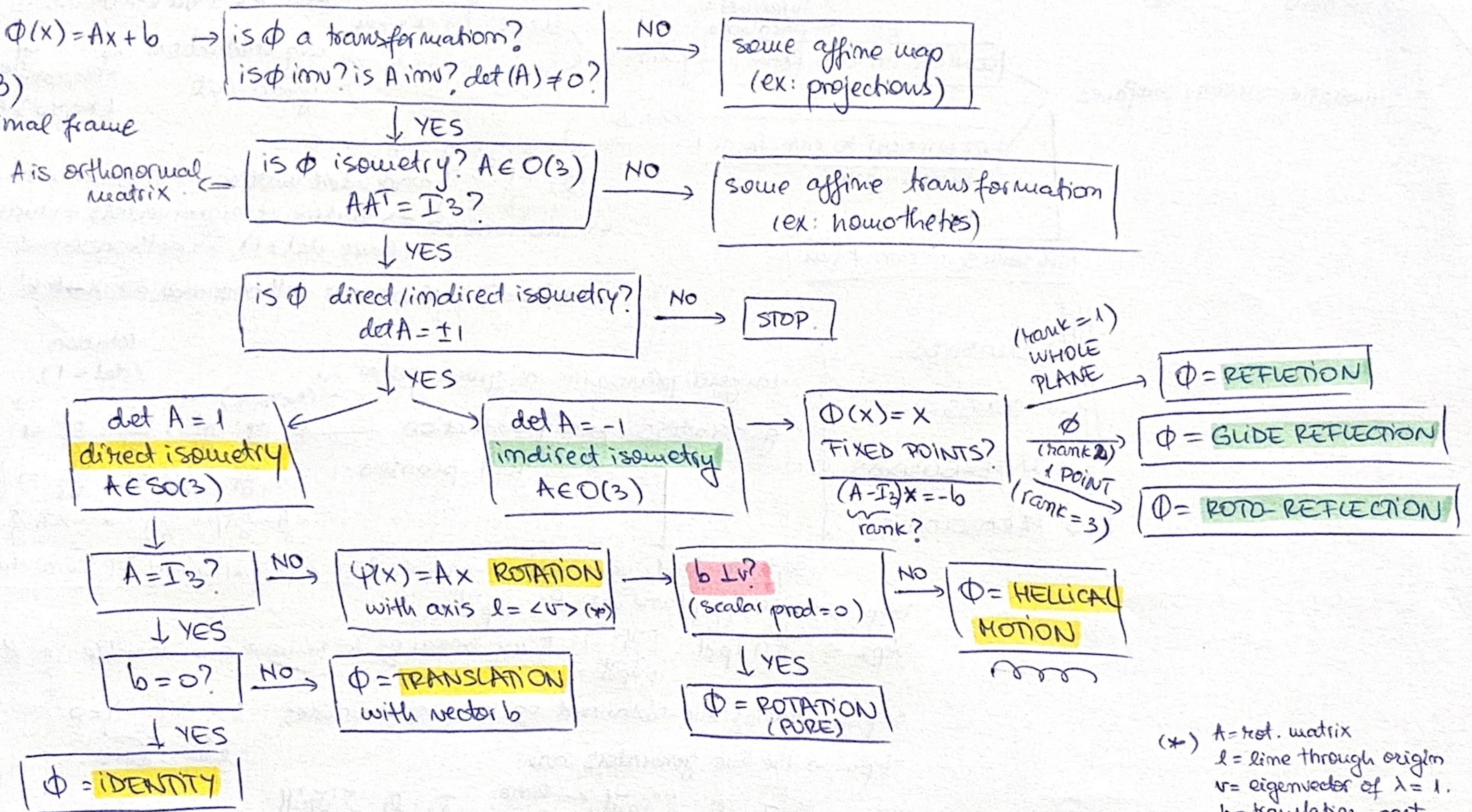
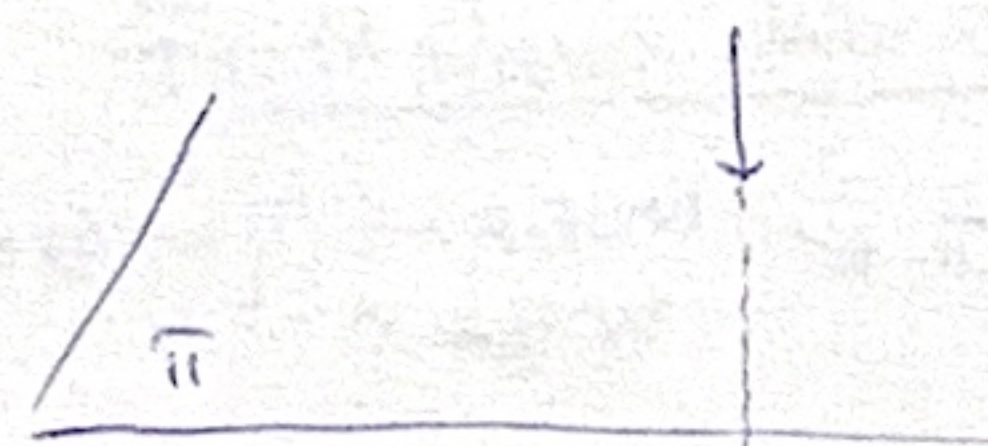


$\Phi: \mathbb{E}^3 \rightarrow \mathbb{E}^3, \Phi(x) = Ax + b$
 $K = (O, B)$
 orthonormal frame



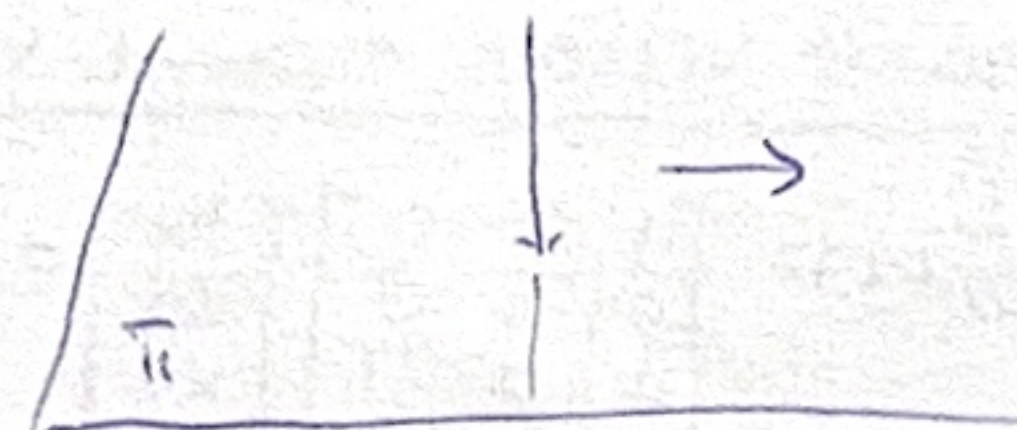
(*) A = rot. matrix
 l = line through origin
 v = eigenvector of $\lambda = 1$.
 b = translation part

FIXED POINTS: reflection



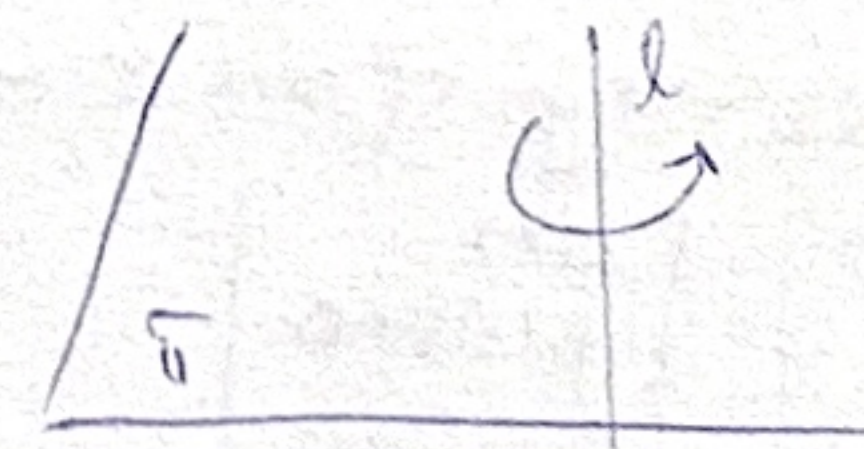
every point is fixed
 $\Rightarrow$ fixed points = $\bar{\pi}$

glide-reflection



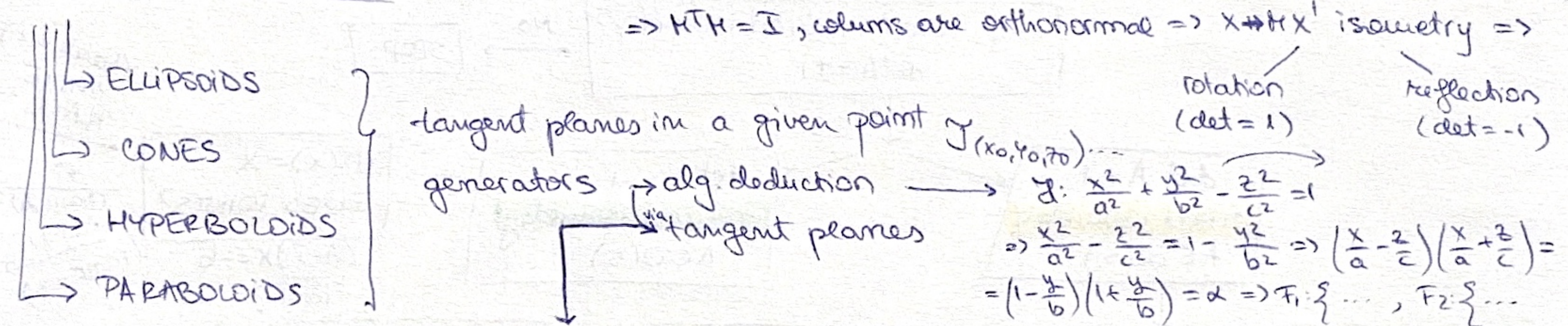
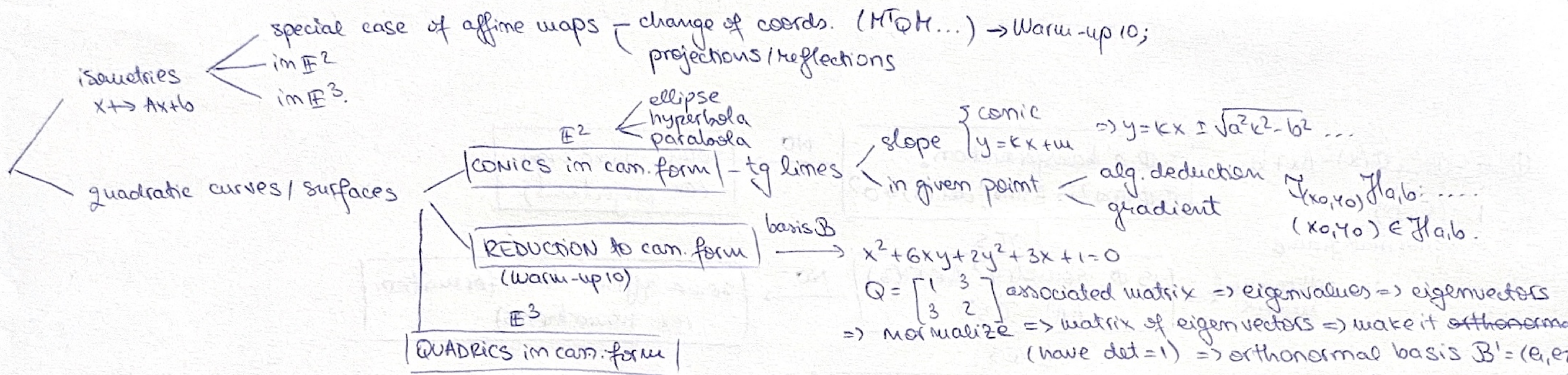
every point moves
 $\Rightarrow$ fixed points = $\emptyset$

rot-reflection



every point on the axis l is fixed
 $\Rightarrow$ fixed points = $\bar{\pi} \cap l$

If $b \cdot v = 0 \Rightarrow$ translation part is perpendicular to the axis around which we rotate ($b \perp v$)
 If $b \cdot v \neq 0 \Rightarrow$ sliding along the axis $\Rightarrow$ helical motion
 $\Rightarrow$ pure rotation.



Steps to find generators in point P of a hyperboloid $\mathcal{H}$ (via tangent plane):

- Step 1: $\rightarrow$ Write tg plane at P: $\mathcal{T}_P \mathcal{H}$
- Step 2: $\rightarrow \mathcal{H} \cap \mathcal{T}_P \mathcal{H} : \begin{cases} \mathcal{H}: \dots \\ \mathcal{T}_P \mathcal{H}: \dots \end{cases} \rightarrow$ use linear eq. to remove one variable in the eq. of $\mathcal{H}$.
- Step 3: $\rightarrow$ Factor the obtained eq. in two variables: $\underbrace{(\quad)}_{\lim_1} \cdot \underbrace{(\quad)}_{\lim_2} = 0$
- Step 4: $\rightarrow$ The two generators are:
 $\mathcal{F}_1: l_1: \begin{cases} \mathcal{T}_P \mathcal{H} \\ \lim_1 = 0 \end{cases}, \mathcal{F}_2: l_2: \begin{cases} \mathcal{T}_P \mathcal{H} \\ \lim_2 = 0 \end{cases}$

Rotate hyperbola around x-axis $\Rightarrow$ hyp. of one sheet:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \mapsto \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}}_{\text{rotation matrix (elliptical motion)}} \underbrace{\begin{bmatrix} \alpha(t) \\ 0 \\ \beta(t) \end{bmatrix}}_{\text{param. of a line}} = \begin{bmatrix} \alpha(t) \\ -\beta(t) \sin \theta \\ \beta(t) \cos \theta \end{bmatrix}$$

Rotate vectors around x-axis with angle $\theta = \pi$

$\text{Ref}_{Ox}^{\perp}(x) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \text{Rot}_{Ox, \pi} \leftarrow \begin{cases} \cos \theta = -1 \\ \sin \theta = 0 \end{cases} \Rightarrow \theta = \pi$

Ox-axis stays the same, $\det A = 1 \Rightarrow$ rotation, not reflection.

Ex: $\mathcal{H}: \frac{x^2}{1} + \frac{y^2}{1} - \frac{z^2}{1} = 1$
 $\mathcal{B} = (1, 0, 0) \in \mathcal{H}$
 $\mathcal{T}_P \mathcal{H}: x x_0 + y y_0 - z z_0 = 1$
 $\mathcal{T}_P \mathcal{H}: x = 1$
 $\begin{cases} x^2 + y^2 - z^2 = 1 \\ x = 1 \end{cases} \rightarrow y^2 - z^2 = 0$
 $\Rightarrow (y-z)(y+z) = 0$
 $\mathcal{I}: \Rightarrow y = z, x = 1$
 $\Rightarrow (x, y, z) = (1, t, t)$
 $\mathcal{II}: \Rightarrow y = -z, x = 1$
 $\Rightarrow (x, y, z) = (1, -t, t)$